

Anirudh Garg

✉ ag9563@nyu.edu • [in anirudhgarg2801](https://www.linkedin.com/in/anirudhgarg2801) • [anirudh28](https://github.com/anirudh28)
🌐 anirudh28.github.io

Education

New York University - Courant Institute of Mathematical Sciences

Aug 2023 – May 2025

M.S. in Computer Science

Indian Institute of Technology, Kanpur

July 2017 – May 2021

B.Tech. in Electrical Engineering

Technical Skills

- **Courses:** Big Data Processing, Deep Learning, NLP, Machine Learning, Distributed Systems, Algorithms
- **Programming Skills:** Python, PyTorch, SQL, Spark, Hadoop, MapReduce, Trino, C++, CUDA, GPU, Git

Work Experience

- **Software Engineer - Google, Mountain View, CA, USA** (Oct '25 – Present)
 - Optimized the **Learned Ranking Function** for **YouTube Shorts** by engineering new behavioral features, resulting in increased user retention and improved content discovery across the highly diverse global platform
 - Conducted rapid A/B testing on ranking models, refining the balance between exploration and exploitation.
- **Software Engineer - Amazon, Dallas, TX, USA** (June '25 – Oct '25)
 - Implemented migration of internal pipelines from **JDK 8 to JDK 17**, which improved performance, resolved backward compatibility issues, managed dependencies, and addressed runtime changes across services
 - Refactored codebase to adopt **JDK 17 features**, and coordinated with teams for a smooth production rollout
- **Machine Learning Engineer - Samsung Research, Bangalore, India** (July '21 – July '23)
 - Developed a **few-shot** learning classification framework to assess Bixby Voice Assistant's performance across **35 different capsules**, classifying user sessions using **data augmentation** techniques and **attention** models
 - Engineered unified attention model, reducing processing time by **97%** and achieving **90%** classification accuracy
- **Summer Intern - Samsung Research, Bangalore, India** (May '20 – July '20)
 - Trained and optimized multiple recommendation models for the Bixby Voice Assistant with an accuracy of **87%**
 - Benchmarked performances of the classification models for the recommendation system for Bixby Voice Assistant

Research Publication

- A Semi-Supervised Approach for Multi-Domain Classification - **Anirudh Garg**, Kartikey Singh, Radhika Mundra (Accepted and published) [\[ICEET 2023\]](#)

Projects

- **EVALUATING LARGE LANGUAGE MODELS FOR EFFICIENT QA SYSTEMS** (Sept '24 – Dec '24)
Mentor: Prof. He He - Computer Science Department, NYU CDS [\[Report\]](#) [\[Code\]](#)
 - Conducted a detailed study of BERT, T5 and Mamba models on QA tasks, balancing EM scores and efficiency
 - Explored advanced tuning techniques like LoRA (Low Rank Adaptation), Quantization and QLoRA for T5, achieving a **75%** reduction in memory usage and **60%** drop in training time with minimal drop in EM score
- **DISTRIBUTED FINANCIAL RISK ASSESSMENT** (Sept '24 – Dec '24)
Mentor: Prof. Yang Tang - Computer Science Department, NYU Courant [\[Report\]](#) [\[Code\]](#)
 - Developed a scalable financial risk assessment system using **Apache Spark MLlib** and **Monte Carlo** simulations
 - Simulated **100,000** portfolio return trials, quantified risks, and delivered robust decision-making financial metrics
- **EMERGENCY RESPONSE EFFICIENCY ANALYSIS** (Sep '24 – Dec '24)
Mentor: Prof. Yang Tang - Computer Science Department, NYU Courant [\[Report\]](#) [\[Code\]](#)
 - Analyzed San Francisco fire and EMS response times, using **HDFS MapReduce** and **Trino** to improve safety
- **FUTURE VIDEO FRAME SEGMENTATION PREDICTION** (Sept '23 – Dec '23)
Mentor: Prof. Yann LeCun - Computer Science Department, NYU Courant [\[Report\]](#) [\[Code\]](#)
 - Employed SimVP and Vision Transformer models with diverse object attributes to predict the segmentation
 - Used the first 11 frames of a video to achieve accurate prediction for the 22nd frame with an accuracy of **40%**
- **VARIABLE SIZED-BATCH GENERAL MULTIPLICATION (VARIABLE GEMM)** (Sept '23 – Dec '23)
Mentor: Prof. Jinyang Li - Computer Science Department, NYU Courant [\[PPT\]](#) [\[Code\]](#)
 - Improved load-balancing in Mixture-of-Experts using CuBLAS and CuSparse, enhancing GPU/TPU efficiency
 - Implemented MoE layer with CuBLAS for non-uniform GEMM, achieving faster, parallelized computations

Scholastic Achievements

- Conferred with a merit certificate at **2016 Indian National Math Olympiad**, ranked among top 50 students
- Awarded with the prestigious **KVPY** fellowship in 2016 with an **All India Rank 370** among 40,000 students

Scholastic Achievements

- Cleared the **Software Competency Test - Professional Level** within 5 months of joining the Samsung Institute
- Secured **All India Rank 799** in **JEE Advanced 2017** among the 220,000 candidates who were shortlisted
- Conferred with a certificate of merit at **Indian National Mathematical Olympiad in 2016** after clearing the **Regional Mathematical Olympiad(North West Region) in 2014 and 2015** with ranks within the top 10%
- Cleared **National Standard Examination in Chemistry 2016-17** and qualified for **Indian National Chemistry Olympiad** used for selecting the Indian Team for International Chemistry Olympiad
- Finished in **National Top 1%** in **National Standard Examination in Physics 2016-17** among 44000 candidates
- Cleared a **2-Tier** selection process of **KVPY 2016** with an **All India Rank 370** among 40,000 students

Relevant Courses

- Advanced Topics in Machine Learning, Machine Learning in Wireless Systems and Data Mining
- Data Structures and Algorithms, Fundamentals of Computing and Image Processing
- Probability and Statistics, Differential Equations, Linear algebra, and Complex Variables and Analysis

Technical Skills

- **Programming Languages:** C, C++, Python, SQL, R, MATLAB, Rust, HTML, JavaScript
- **Related Libraries:** Sklearn, Tensorflow, Pandas, NumPy, SciPy, OpenCV, Pytorch
- **Operating Systems and Softwares:** LINUX, Windows, Anaconda, GNU Octave, MacOS, Arduino IDE

Positions of Responsibility

Student Guide, Counselling Service, IIT Kanpur

(July'18-April'19)

- Helped in organizing workshops and sessions for the incoming batch of around **900 students** during **Orientation 2018** ensuring their smooth acclimatization to the campus surroundings
- Assisted six first year students emotionally and informatively, ensuring their smooth transition to college life

Secretary, Electronics Club, IIT Kanpur

(July'18-April'19)

- Delivered lectures on IOT, communication and Logic Circuits during the winter workshop in December'18 held by the club
- Conducted electronics events of inter-hall technical competition TAKNEEK'18 with a team of 20 members
- **COMMUNICATION ARCHITECTURE FOR UAV SWARMS** (Summer Project) (May '19 – July '19)
Mentor: Prof. Ketan Rajawat - Electrical Engineering Department, IIT Kanpur [\[Code\]](#)
 - Executed decision-making for **UAV** communication between Wireless ad-hoc Network and RFD900X
 - Conceived interface bridging to route Ethernet-collected data wirelessly, significantly extending the communication range from 1 km to 40 km, thus enabling the communicating with the UAV's over longer distances